

Solution Requirements

Date	15 March 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

FR ID	Functional Requirement	Description	Priority
FR-01	User Registration	Citizens and officials can register in the system.	High
FR-02	User Login	Users can securely log in to access services.	High
FR-03	Report Flood Incident	Users can report flood-related incidents with location and details.	High
FR-04	View Flood Alerts	Users can view real-time flood alerts and warnings.	High
FR-05	Track Water Levels	The system displays current water level information.	High
FR-06	Emergency Notifications	The system sends emergency alerts to registered users.	High
FR-07	View Safe Shelters	Users can locate nearby emergency shelters.	High
FR-08	Admin Management	Administrators can manage users, reports, and alerts.	High
FR-09	Generate Reports	The system generates flood analysis and response reports.	Medium
FR-10	User Logout	Users can securely log out of the application.	High

Step 2: Non-Functional Requirements (NFR)

NFR ID	Non-Functional Requirement	Description	Priority
NFR-01	Performance	The system shall respond to user requests within 3 seconds under normal operating conditions.	High
NFR-02	Availability	The system shall be available 24/7, except during scheduled maintenance.	High
NFR-03	Reliability	The system shall operate consistently with minimal downtime and accurate results.	High
NFR-04	Security	User data shall be protected using authentication, authorization, and encryption.	High
NFR-05	Scalability	The system shall support an increasing number of users and data without performance degradation.	High
NFR-06	Usability	The user interface shall be simple, intuitive, and easy to navigate.	Medium
NFR-07	Maintainability	The system shall be easy to maintain, update, and fix when required.	Medium
NFR-08	Compatibility	The system shall work on major web browsers and mobile devices.	Medium
NFR-09	Backup and Recovery	The system shall perform regular data backups and support recovery after failures.	High
NFR-10	Compliance	The system shall comply with applicable data privacy and security regulations.	Medium